Just IT

B2W group

Apprenticeships | Training | Recruitment

Data Technician

Name: Jonathan Akpofure

Course Date: 19/1/26

Table of contents

Day 1: Task 1 2
Day 2: Task 1 2
Day 2: Task 2 3
Day 2: Task 3 4
Day 3: Task 1 4
Day 3: Task 2 5
Dataset: 5
Step 1: Create a Pivot Table 5
Step 2: Use the SWITCH Function 5
Submission: 6
Day 3: Task 3 6
Day 4: Task 1 7
Course Notes 9
Additional Information 10

Day 1: Task 1

Please complete the below boxes on commons laws and regulations that must be followed when working with customers data, use the below bulleted list to support your answers.

- What is it
- Why is it important
- Provide a real-world example of how you can follow it
- How does it impact working with data
- What could happen if you breached it

Data Protection Act	The Data Protection Act is a UK Law that controls how businesses use your personal information (names, addresses, etc.). The Data Protection Act is important because it helps to prevent cybercrime and allows for quick recovery of data after incidences.

	A real-life example of how you can follow the data protection act is by locking your screen once you're away from your desk and by not using public Wi-Fi when handling important matters/tasks. The Data Protection Act impacts working with data by requiring businesses to handle personal data fairly and honestly by ensuring security and consent to not face hefty fines. If you breach the Data Protection Act, then you could get official warnings from the data protection authorities and ban on processing data
GDPR	GDPR is a strict EU Law that gives people more control over their personal data. GDPR is important because it builds customer trust through transparency and honesty. A real-life example of how you can follow the GDPR could be an agency that realizes an employee accidentally clicked on a malicious web address in a scam message; exposing customer email addresses. Within a couple days the agency gets a hold of the Information Commissions Office to notify them of what happened, and they inform the individuals affected of what occurred. GDPR Law impacts working with data by needing consent to handle and process the data. You also clearly inform the individuals how you plan to use their personal data (right to inform) and be able to show compliance. One consequence could be massive fines. Information Commissions Office can issue fines up to 20million euros. Another consequence could be reputational damage where customer trust could be worsened which could lead to negative publicity
Freedom of Information Act	Freedom of Information Act is an act that gives everybody the right to request recorded info like medical details on the computer system. This act is important because it helps with public trust as everyone has the right to their recorder info. It's also important as it encourages accountability and honesty from the government. A real-life example of how to follow the freedom of information act could be data on local recreation. You believe there are not many activities for children, and you want to know how much the government spends on recreational activities for the local children. You send an email to the council making a request. Within 20 days, you get an email back from the council showing the money allocated to recreational activities. You

	compare the money allocated to how many activities are available for the kids. You use this information to start a community campaign. The Freedom of Information Act impacts working with data as businesses/organizations must support a publication strategy by regularly making data readily available. Also, it impacts working with data as due to the act responses must be sent back within 20 days; which means businesses must have a recoverable management system. If you are found to have breached the Freedom of Information Act, this could lead to complaints to the Information Commissioner's Office (ICO) which as a result could mean penalty notices for any major breaches.
Computer Misuse Act	The Computer Misuse Act is an act/law that stops hackers from doing unethical practices with data that isn't theirs through hacking. It's important because it helps to protect computer systems by deterring cybercrime. A real-life example of how you can follow the computer misuse act is by using a secure and strong password in the workplace to ensure only certain employees have access to the files and systems; making sure to step away from your desk once the screen is locked. The computer misuse act impacts working with data as you can't copy someone's data without permission as it is illegal. This helps to prevent hacking and fraud. If you break the act, you can face significant prison time which could lead to having a criminal record.

Day 2: Task 1

Please research and complete the following tasks within the retail-sales_dataset.xlsx document, paste a print screen into the provided boxes below:

1. In the sheet 'retail_sales_dataset' add all available data between columns A –J into a 'table'
2. Using the 'sort' function, sort 'Age' to 'largest to smallest'
3. Using the 'SUM' function, show me the commission total in cell 'L10'

4. Using the ‘AVERAGE’ function, show me the average commission in cell ‘L11’

Print screen 1

File Home Insert Share Page Layout Formulas Data Review View Help Draw Table Design

PivotTable Table Forms Pictures Shapes Checkbox Slicer Link New Comment

Female

	A	B	C	D	E	F	G	H	I	J
1	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total_Sales	Commision 2023
2	14	1/17/2023	CUST014	Male	64	Clothing	4	£ 30.00	£ 120.00	£ 1.80
3	25	12/26/2023	CUST025	Female	64	Beauty	1	£ 50.00	£ 50.00	£ 0.75
4	80	12/10/2023	CUST080	Female	64	Clothing	2	£ 30.00	£ 60.00	£ 0.90
5	122	10/3/2023	CUST122	Male	64	Electronics	4	£ 30.00	£ 120.00	£ 1.80
6	161	3/22/2023	CUST161	Male	64	Beauty	2	£ 500.00	£ 1,000.00	£ 15.00
7	163	1/2/2023	CUST163	Female	64	Clothing	3	£ 50.00	£ 150.00	£ 2.25
8	173	11/8/2023	CUST173	Male	64	Electronics	4	£ 30.00	£ 120.00	£ 1.80
9	187	6/7/2023	CUST187	Female	64	Clothing	2	£ 50.00	£ 100.00	£ 1.50
10	191	10/18/2023	CUST191	Male	64	Beauty	1	£ 25.00	£ 25.00	£ 0.38
11	218	9/22/2023	CUST218	Male	64	Beauty	3	£ 30.00	£ 90.00	£ 1.35

Print screen 2

E
Age
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64

Print screen 3

=SUM(J2:J1001)

J			K	L		M
es	Commision 2023					
0.00	£	1.80		Column1		Column2
0.00	£	0.75				rate %
0.00	£	0.90		Commission 2023		1.50%
0.00	£	1.80		Commission 2024		2.00%
00.00	£	15.00				
0.00	£	2.25				
0.00	£	1.80				
00.00	£	1.50				
5.00	£	0.38		£	6,840.00	Total
0.00	£	1.35		£	6.84	Total

Print screen 4

=AVERAGE(J2:J1001)

J			K	L		M
s	Commision 2023					
0.00	£	1.80		Column1		Column2
0.00	£	0.75				rate %
0.00	£	0.90		Commission 2023		1.5
0.00	£	1.80		Commission 2024		2.0
0.00	£	15.00				
0.00	£	2.25				
0.00	£	1.80				
0.00	£	1.50				
5.00	£	0.38		£	6,840.00	Total
0.00	£	1.35		£	6.84	Total

Day 2: Task 2

Please research and complete the following tasks within the retail-sales_dataset.xlsx document, paste print screens into the provided box below:

Student name	English	Mathematic:	Science	Average	Highest score
Carol	75	85	85		
Ted	80	75	90		
Khan	85	75	80		
Harry	80	70	80		
Sarah	80	70	80		
John	65	80	70		
Linda	90	50	70		
Edward	55	80	60		
Mary	55	70	65		
Thomas	55	30	65		

Task

1) Apply filter and sorting to show the best students in each subject.
2) Calculate the average for all students and fill into Column E. (Use formula)
3) Using the =MAX fucntion, tell me what the students highest score was in column F.
4) Apply filter and sorting to show the best student in this classroom by average.
5) Apply filter and sorting to show the best student in this classroom by highest score.
6) Use conditional formatting to clearly identify the highest and lowest average scores

Print screen 1

1)

	A	B
1	Student name	English
2	Linda	90
3	Khan	85
4	Ted	80
5	Harry	80
6	Sarah	80
7	Carol	75
8	John	65
9	Edward	55
10	Mary	55
11	Thomas	55

	A	B	C
	Student name	English	Mathema
	Linda	90	85
	Khan	85	80
	Ted	80	80
	Harry	80	75
	Sarah	80	75
	Carol	75	70
	John	65	70
	Edward	55	70
	Mary	55	50
	Thomas	55	30

A	B	C	D
Student name	English	Mathema	Scienc
Ted	80	80	90
Carol	75	70	85
Khan	85	80	80
Harry	80	75	80
Sarah	80	75	80
Linda	90	85	70
John	65	70	70
Mary	55	50	65
Thomas	55	30	65
Edward	55	70	60

E
Average
81.666667
65
76.666667
71.666667
80
70
63.333333
76.666667
81.666667
50

2)

F
Highest score
90
85
85
80
80
80
90
70
65
80

3)

4)

	A	B	C	D	E
	Student name	English	Mathema	Scienc	Average
	Carol	75	85	85	81.666667
	Ted	80	75	90	81.666667
	Khan	85	75	80	80
	Harry	80	70	80	76.666667
	Sarah	80	70	80	76.666667
	John	65	80	70	71.666667
	Linda	90	50	70	70
	Edward	55	80	60	65
	Mary	55	70	65	63.333333
	Thomas	55	30	65	50

	A	B	C	D	E	F
	Student name	English	Mathema	Scienc	Average	Highest score
	Ted	80	80	90	83.333333	90
	Linda	90	85	70	81.666667	90
	Carol	75	70	85	76.666667	85
	Khan	85	80	80	81.666667	85
	Harry	80	75	80	78.333333	80
	Sarah	80	75	80	78.333333	80
	John	65	70	70	68.333333	70
	Edward	55	70	60	61.666667	70
	Mary	55	50	65	56.666667	65
	Thomas	55	30	65	50	65

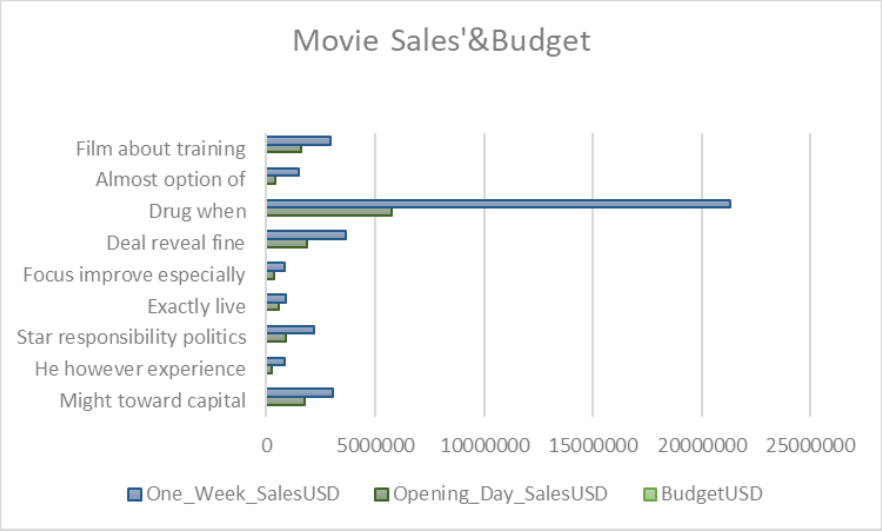
Average
61.666667
56.666667
50
68.333333
76.666667
83.333333
78.333333
78.333333
81.666667
81.666667

Day 2: Task 3

Using the skills developed today, have some fun with the data set you have imported. Paste your work below and enjoy!

Print screen 1

Title	Genre	BudgetUSD	Opening_Day_SalesUSD	One_Week_SalesUSD	If
Might toward capital	Comedy	6577427.79	1778530.85	3034053.32	
He however experience	Comedy	1883810.1	247115.74	831828.84	
Star responsibility politics	Comedy	2468079.29	878453.95	2171405.93	
Exactly live	Comedy	1447311.46	570657.72	898886.01	
Focus improve especially	Documentary	900915.86	361189.37	861775.91	
Deal reveal fine	Drama	6356058.11	1877251.88	3681524.22	
Drug when	Comedy	30881640.04	5771559.63	21305870.04	
Almost option of	Horror	5249748.92	429972.59	1513066.66	
Film about training	Drama	4746218.15	1612321.12	2977114.88	



Day 3: Task 1

Please download the dataset 'Day_3_Task_1_Bike_Sales_Pivot_Lab.xlsx' and the lab instructions.

Do not worry if you do not complete the lab, just working with data and playing with the pivot table will be good experience.

Please paste your final pivot table below and complete the reflection questions:

Print screen 1	

Sum of Order_Quantity		Country								
Age_Group	Customer_Gender	Australia	Canada	France	Germany	United States	United Kingdom	United States	United States	Grand Total
Youth (<25)	F	9	0	6	0	0	1	0	0	16
	M	2	0	4	0	0	5	0	0	11
Youth (<25) Total		**11**	**0**	**10**	**0**	**0**	**6**	**0**	**0**	**27**
Young Adults (25-34)	F	17	6	1	0	0	3	10	0	37
	M	3	5	9	0	0	1	6	0	24
Young Adults (25-34) Total		**20**	**11**	**10**	**0**	**0**	**4**	**16**	**0**	**61**
Adults (35-64)	F	17	0	0	8	2	1	27	0	55
	M	15	0	0	5	0	3	20	1	44
Adults (35-64) Total		**32**	**0**	**0**	**13**	**2**	**4**	**47**	**1**	**99**
Grand Total		**63**	**11**	**20**	**13**	**2**	**14**	**63**	**1**	**187**

In which markets does Germany have customers?	Adults (35-64)
What country has sales in all markets?	UK and Australia
What are the most profitable markets by country, age group, and gender?	Country- United States Age Group- Adults (35-64) Gender- Female
Any other findings?	Germany only has sales from adults (35-64). No sales from young adults (25-34) and youth (<25). France has customers for young adults and youth but not for adults.

Day 3: Task 2

The dataset below tracks the sales performance of different products in various counties in England. Please paste the dataset into a blank Excel workbook. Your task is to:

- **Create a Pivot Table** to summarise the data by county and product.
- **Use the SWITCH function** to categorise products based on their sales volume.

Dataset:

County	Product	Sales Volume
Yorkshire	Laptops	500
Yorkshire	Smartphones	200
Cornwall	Laptops	700
Cornwall	Printers	400
Lancashire	Smartphones	150
Lancashire	Laptops	600
Essex	Printers	800
Essex	Smartphones	300
Durham	Laptops	250
Durham	Printers	300
Greater Manchester	Smartphones	600
Greater Manchester	Laptops	400

Step 1: Create a Pivot Table

- Select the dataset (columns A to C).
- Insert a Pivot Table to summarise the data by **County** in the rows and **Products** in the columns. Use **Sales Volume** as the value to be summarised.

Step 2: Use the SWITCH Function

In a new column next to your data, use the SWITCH function to categorise products based on **Sales Volume** as follows:

- For sales greater than 600: **"High"**
- For sales between 300 and 600: **"Medium"**
- For sales less than 300: **"Low"**

SWITCH Function Example:

```
=SWITCH (TRUE, C2 > 600, "High", C2 >= 300, "Medium", "Low")
```

- Apply this formula to each row, and check if the products are categorised correctly.

Submission:

- A completed Pivot Table summarising sales by county and product.
- A new column in the dataset categorising products by sales volume using the SWITCH function.
    - Please paste your completed work below

Print screen 1

```
=SWITCH(TRUE, C2 > 600, "High", C2 >= 300, "Medium", "Low")
```

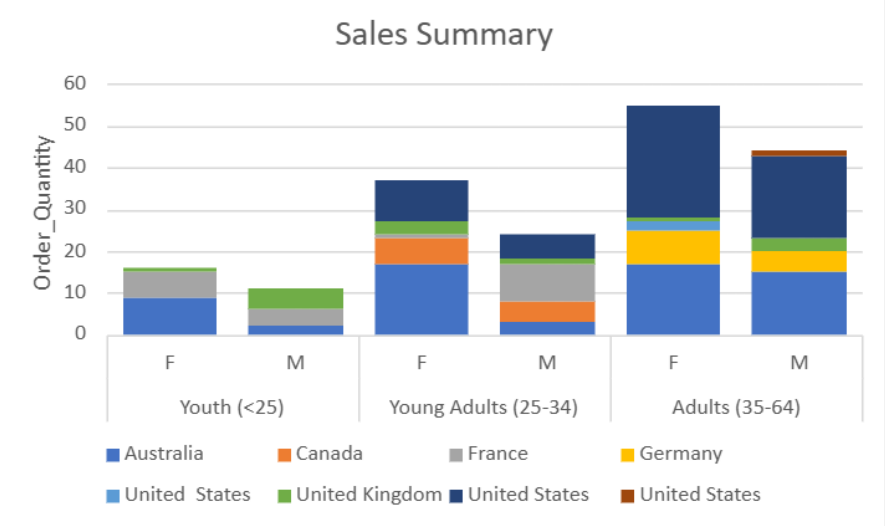
Sum of SALES VOLUME	PRODUCT					
COUNTY	Laptops	Printers	Smartphones	Smartphones	Grand Total	
Yorkshire	500				500	Low
Cornwall	700	400			1100	Low
Durham	250	300			550	Low
Essex		800	300		1100	Low
Greater Manchester	400			600	1000	Low
Lancashire	600		150		750	Low
Yorkshire			200		200	Low
Grand Total	**2450**	**1500**	**650**	**600**	**5200**	

Day 3: Task 3

Please download the dataset 'Day_3_Task_3_Bike_Sales_Visualisations_Lab.xlsx' and the the lab instructions. Do not worry if you do not complete the lab, just working with data and playing with the charts will be good experience.

Please paste your results below:

Print screen 1



Day 4: Task 1

You have been asked to deliver your analysis findings to the board of directors, with your analysis you have identified that customers are leaving your company at the 12-month point, this is typically when they receive their renewal price.

Conduct research and complete the below questions:

How would you prepare for the delivery?	I would prepare to deliver my findings to the board by firstly arriving early. I would meet with the main board members in advance to understand their expectations and current knowledge on the matter at hand. For any important decisions that can't wait before the delivery, I would speak with key members, give my own recommendation and then they can share their thoughts. In my report, I would prepare a summary page or two highlighting the main points and any risks involved. I would make charts with the data to make it easy to understand as they're non-data-analysts and make a colour-coded table to make the data easier to know. I would create a small slideshow presentation to show patterns, to then have an accurate prediction of what things may look like soon.
What tools would you use for the delivery?	Tableau Power BI Excel PowerPoint

What is prospecting and why would you complete this before your delivery?	Prospecting is talking to key stakeholders prior to the meeting. You would do this before your delivery to understand their own goals and any concerns they may have to tailor your findings to their specific issues.
Tell me best practices for public speaking and providing updates to senior leaders	Public Speaking and Providing updates to senior leaders- Pause for a few seconds before you start speaking to get the room's attention. Stop for a few seconds after a big/main point, to signal more important information is coming Speak for 15 mins &Answer questions for 15 mins
What will you show the board in your delivery?	Core/Main Findings Major Risks involved Honest view on how the company is performing currently and what it could look like in the next 5 years using data Show what the data (graphs, charts) means/indicates Show the best case, worst case and most likely case to make it easier for the board to plan the next course of action (Giving your recommendations)

How will you articulate the changes that are needed?	Start with your own recommendation immediately in a clear precise manner Describe what the organisation will look like after the change Be precise about what is changing, what is not changing and what will not be a priority Present a roadmap highlighting the major parts Identify potential issues and outline plants to mitigate them Define measurable outcomes/KPI'S to monitor progress to ensure that the change is working Leave time for Question and Answer
Provide a list of online resources and videos that will support your preparation for public speaking	Poised Communication Coach Alex Lyon (YouTube) MasterTalk(YouTube) Charisma on Command(YouTube) Dynamic Public Speaking Specialisation(University of Washington) - Online Course

Evaluate tools that provide visualisation. **Tell me what they are.** **Tell me what you would choose when delivering your presentation and why**	Power BI Tableau Excel These are the main tools for data visualisation Power BI- Power Query engine handles complex tasks, while the DAX engine helps with advanced data calculations. Affordable and has many customisable actions which allows for more interactivity Though it's hard to master DAX for complex situations as it requires a lot of effort. Tableau- Connects to many data sources. Maintains strong security and can handle large datasets Though it's very expensive and standard versions doesn't have an automatic refresh for your data Excel- Wide range of charts and can create interactive dashboards by combining Pivot Charts, Pivot Tables allowing you to explore data smoothly Though it may be slow/inefficient when handling large and complex datasets. Also I would choose Power BI because it combines AI Powered storytelling, with live PowerPoint integration and interactive dashboard which allows you to answer board questions easily

Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.